

Project Proposal

Data Visualization 2025–2026 · University of Genova

Student: Mojtaba Alehosseini

Format: Solo project

What the project is

The site is called "The Price of Europe" and tells the story of Europe's cost of living crisis from 2019 to 2025. It's a single-page scrollytelling experience, the reader scrolls through five chapters: the pre-crisis baseline, the COVID shock, the energy crisis after Russia's invasion of Ukraine, food inflation cascading from energy costs in 2023, and where things stand now. The idea is to go beyond "prices went up" and show which prices, in which countries, and whether wages actually kept up.

Data

All data comes from Eurostat. The datasets cover:

- Monthly inflation rates and price index levels (HICP), 2018–2025
- Household electricity prices, by semester
- Statutory minimum wages across EU countries
- House price index, quarterly

A Python/Jupyter notebook handles all preprocessing and will be included in the repository with a requirements.txt and run instructions.

Tech stack

HTML, CSS, and JavaScript with D3.js v7. No frameworks. Deployed to GitHub Pages. Scrollytelling is handled by Scrollama.

Visualizations

The plan is eight core charts, each with at least one non-trivial interaction or computation beyond a basic tooltip:

- Animated choropleth, EU inflation map with a play/pause timeline and click-to-detail per country
- Small multiples, inflation by product category, with cross-panel country highlighting
- Heatmap, country vs. category matrix with sort toggle and time scrubber
- Slope chart, electricity prices 2019 vs. 2024
- Annotated line, Euro Area inflation timeline with event markers (Ukraine war, ECB rate hikes, etc.)
- Stacked area, inflation drivers over time, switchable between stacked, streamgraph, and normalized views
- Diverging bar, real wage change per country (wage growth minus inflation)
- Waffle chart, what €100 from 2019 buys today, switchable by country and category

Up to seven more charts are planned as an optional extension, to be built only after the core eight are fully working and deployed.

Design and structure

Dark theme, single-page app. Typography: Roboto Slab for headings, Fira Sans for body text. The page has a header, a hero section that frames the story, five narrative sections, a methodology section, and a footer.

Methodology section

The site will include a visible methodology section with data sources and links, the preprocessing pipeline, known limitations (for example: five EU countries have no statutory minimum wage, and Eurostat data has a reporting lag of a few months), and notes on how uncertainty is shown in the charts.

Deliverables

- Public GitHub repository with clean structure and commit history
- Live site on GitHub Pages
- Preprocessing notebook, fully runnable and documented
- README with folder structure, local server instructions, and data sources